SAI KALYANDURG

Senior Java Full Stack Developer

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PROFESSIONAL SUMMARY

Java full stack engineer and engineering manager with 5 years of experience designing, building and leading enterprise systems across capital markets, healthcare and retail. Core expertise in Java 21, Spring Boot, microservices and event-driven architecture with Apache Kafka, paired with React frontends and production delivery on AWS and GCP using Docker, Kubernetes, Terraform and GitOps CI/CD. Most recently managed a CFTC-regulated books and records migration at ADM Investor Services, moving up to 15 years of FCM data from a legacy VAX platform to Microsoft Dynamics 365 in 7 to 9 months against a 12 to 18 month plan, while building low-latency FIX market data integrations for its trading platforms. At Kintsugi, led a team building HIPAA-compliant, API-first clinical AI services with real-time WebSocket scoring and HL7 FHIR integrations, and at Lowe's co-architected a k6-based performance engineering platform that cut infrastructure cost by 60% for 40+ teams ahead of Black Friday. Brings depth across the full software lifecycle, from system design, API governance and data migration to observability, JVM tuning, security and compliance, and is comfortable moving between architecture, hands-on code and team leadership.

WORK EXPERIENCE

ADM Investor Services | Chicago, IL

Manager ***Dec 2025 – Aug 2026***

- Managed the ADMIS UK to ADMIS US books and records migration for a CFTC-registered futures commission merchant (FCM), moving 10 to 15 years of customer, trade, margin, risk, general ledger and compliance data from a legacy VAX platform to Microsoft Dynamics 365 Finance & Operations, and compressed the program from a 12 to 18 month estimate to 7 to 9 months through wave-based migration, reusable mapping templates and automated reconciliation.
- Finalized the target-state architecture and data flow diagrams, decomposed the program into Jira epics and user stories, and led a 12-member team of engineers, data analysts and QA as hands-on technical lead, reporting to treasury, finance, risk, compliance and UK and US business units through status reviews and RAID logs.
- Built Python and Spring Batch ETL pipelines on AWS (S3, Glue, EKS) to extract legacy VAX fixed-width and indexed files and load them through D365 Data Management Framework and OData entities, migrating 150M+ records with data quality rules that resolved 25K+ issues before cutover.
- Designed source-to-target mappings for customer master, accounts, open positions, cash balances and general ledger journals, preserving segregated (CFTC Regulation 1.20), secured (Part 30) and proprietary fund classifications and supporting net capital (Regulation 1.17), daily segregation statements and Form 1-FR-FCM reporting in D365 without manual adjustments.
- Automated reconciliation with control totals, hash totals, balance-by-account and segregated and secured amount comparisons across 3 mock migrations and a parallel run, achieving a 100% financial balance match, 99.99% record-level accuracy and a zero-disruption weekend cutover.
- Preserved books and records in WORM-compliant S3 Object Lock storage under CFTC Regulation 1.31 and NFA Compliance Rule 2-10, and migrated KYC, CIP and AML records with OFAC history under the Bank Secrecy Act while applying UK GDPR cross-border safeguards and FCA client money (CASS) mapping.
- Enforced SOX-aligned change management and ITGC audit evidence, and secured migration data with AWS Transfer Family SFTP, KMS encryption at rest, Secrets Manager, least-privilege IAM and masked PII in non-production environments, passing internal audit with no findings.
- Drove the .NET to Java (JavaFX) re-platforming of Oak Desktop, a customizable low-latency futures and options trading terminal with real-time electronic quotes, custom waterfall boards and multi-window workspace management, aligning it with the Java-based Oak Web and Nexus Portal stack.
- Developed Oak Web order entry, order tracking, SPAN margin update and trade log export features in React 19 and Spring Boot 3.5, with pre-trade risk checks aligned to CFTC Regulation 1.73 and complete order audit trails, improving order workflow performance by 35%.
- Delivered Nexus Portal modules in Java and Spring Boot giving traders and introducing brokers (IBs) self-service access to daily and monthly statements (CFTC Regulation 1.33), clearing records, capital reports and compliance metrics, reducing client support requests by 30%.
- Integrated TT, CQG, Rithmic and CTS market data and execution feeds through FIX 4.4 with QuickFIX/J, normalizing CBOT and CME Group data onto Amazon MSK with sub-10 ms processing latency, and built FIX session recovery (sequence gap detection, resend requests, auto-reconnect) with dead-letter handling for failed messages, eliminating manual feed recovery.
- Streamed real-time quotes to Oak clients over WebSockets with message conflation and backpressure handling, sustaining 10K+ price updates per second during peak market hours.

- Engineered a SPAN margin service that ingests daily CME SPAN risk parameter files and recalculates account-level initial and maintenance margin, replacing end-of-day batch runs with intraday margin visibility for traders and risk teams.
- Tuned JVM performance for the Java trading stack with ZGC tuning, object pooling and allocation-free hot paths for market data handlers, holding p99 GC pauses under 5 ms during peak market hours, and cached reference data (contracts, accounts, margin rates) in Redis to cut Oak Web response times by 30%.
- Standardized platform engineering across Oak and Nexus with contract-first OpenAPI 3 APIs and URI versioning, global exception handling with RFC 7807 ProblemDetail responses, and externalized configuration through Spring profiles and AWS Secrets Manager, reducing integration and deployment defects by 25%.
- Secured Oak Web and Nexus Portal with Spring Security 6 and OAuth 2.0/OIDC through Microsoft Entra ID with role-based access and CloudTrail audit trails, and added structured logging, correlation IDs and CloudWatch alerting that cut incident triage time by 40%.
- Set up GitHub Actions CI/CD pipelines with Terraform-provisioned infrastructure, code quality gates and Docker deployments to Amazon EKS, with JUnit 5, Testcontainers, FIX simulator and Playwright test stages, moving releases from monthly to weekly while mentoring engineers and leading code reviews.

Kintsugi | Los Angeles, CA
Senior Java Full Stack Developer ***Jan 2024 – Dec 2024***

- Led a team of 6 engineers building Kintsugi Voice, an API-first clinical decision support platform using vocal biomarker AI to detect depression and anxiety from short speech samples, partnering with ML scientists, clinical advisors and product managers to onboard 10+ payer, provider and digital health partners.
- Architected an event-driven microservices platform on AWS EKS with Spring Boot services for audio ingestion, scoring orchestration, triage, scheduling and partner management, connected through REST and Amazon MSK so retrospective and live in-call scoring could scale independently.
- Built scoring APIs in Java 21 and Spring Boot 3.3 that ingest 20 to 30 second speech clips, orchestrate Python FastAPI inference services and return PHQ-9 and GAD-7 aligned severity scores at p95 latency under 2 seconds, with idempotency keys preventing duplicate scoring.
- Engineered real-time in-call scoring with Spring WebFlux and WebSockets using reactive audio streaming, chunking, backpressure and Redis-backed session state, surfacing triage signals to clinicians within 30 seconds during live telehealth sessions.
- Developed a rules-based triage and care routing engine and a React 18 and TypeScript clinician dashboard with role-based access and longitudinal risk trends that flag high-acuity patients and match them to partner provider schedules, reducing time from screening to scheduled care by 60%.
- Built population health scoring pipelines with Spring Batch 5 partitioned jobs and Kafka consumers, processing 500K+ archived call recordings for payer partners to close care gaps among undiagnosed members.
- Integrated results into EHR systems through HL7 FHIR R4 (Patient, Encounter, Observation, QuestionnaireResponse) and SMART on FHIR to embed mental health signals in clinician workflows, and built an HMAC-SHA256 signed webhook service with retries and dead-letter handling for 15+ partners.
- Delivered a developer platform with Java, Python and JavaScript SDKs, a sandbox environment, Postman collections, OpenAPI 3 specifications and versioned partner APIs (/v1, /v2) with deprecation notices and backward compatibility guarantees, reducing partner integration time from ~6 weeks to ~1 week.
- Secured partner APIs with Spring Security 6 using OAuth 2.0 client credentials, scoped JWTs and API Gateway throttling, and implemented HIPAA safeguards (KMS AES-256 encryption, TLS 1.3, least-privilege IAM, audio de-identification, CloudTrail audit trails) supporting HIPAA and SOC 2 Type II compliance.
- Designed a multi-tenant PostgreSQL 16 schema on Amazon RDS with Spring Data JPA, Hibernate and row-level security, versioned through Flyway, and offloaded audio uploads to encrypted S3 presigned URLs with BAA-driven retention, cutting API server load by 50%.
- Standardized exception handling with RFC 7807 ProblemDetail responses and pre-inference audio validation (codec, sample rate, duration, signal-to-noise ratio), cutting failed scoring requests by 40%.
- Established OpenTelemetry tracing across Java and Python services with PHI-redacted structured logs and Datadog SLO alerts, reducing MTTR by 50%, and tuned the JVM with off-heap ByteBuffers and generational ZGC to hold p99 GC pauses under 10 ms.
- Provisioned AWS with Terraform and Helm, autoscaling inference pods on Kafka consumer lag with KEDA to cut compute cost by 35% at 99.95% availability, and shipped through GitHub Actions, SonarQube, Snyk and Argo CD blue-green deployments for multiple production releases per day, managing configuration through Spring profiles and AWS Secrets Manager and canary model rollouts through per-partner feature flags.
- Drove testing with JUnit 5, Mockito, Testcontainers, Pytest and Pact contract tests at 80%+ coverage, and validated 1,000+ concurrent scoring requests with k6 load tests.

Lowe's | Bengaluru, India
Java Full Stack Developer ***Jun 2021 – Jun 2023***

- Led POCs benchmarking JMeter, k6, Gatling and Locust and co-architected Perfex, a multi-tenant load testing platform of four Spring Boot microservices over REST and Kafka, replacing always-on JMeter VMs with ephemeral k6 runners on GKE and scale-to-zero autoscaling to cut infrastructure cost by 60%.
- Enabled shift-left performance testing for 40+ teams, including e-commerce, checkout, order management and inventory, through CI-triggered test runs, surfacing 30+ capacity bottlenecks before peak season and scaling to 3,000+ daily active users ahead of Black Friday.
- Built test lifecycle APIs in Java 17 and Spring Boot 3.0 with a MongoDB-backed state machine and Spring Batch 5 chunk jobs processing 50M+ data points weekly with automatic recovery after pod restarts, and parallelized report queries with CompletableFuture and Caffeine caching for 35% faster responses.
- Developed a k6 script generator with 15 workload templates and built-in P95 thresholds, plus a JMeter-to-k6 converter that migrated 1,000+ .jmx scripts, cutting test authoring time by 40% and team onboarding time by 80%.
- Established contract-first OpenAPI 3 design with URI versioning and Deprecation and Sunset headers, migrating 40+ teams to v2 APIs with zero broken integrations, and added idempotency keys, Bucket4j rate limiting and per-tenant quotas.
- Externalized configuration with Spring profiles, Kubernetes ConfigMaps, Secret Manager and validated @ConfigurationProperties, cutting config-related deployment issues by 70%, and piloted features with 5 teams through per-tenant feature toggles.
- Replaced batch reporting with a Kafka 3.4 event-driven pipeline, reducing report availability from next day to under 5 minutes, and designed MongoDB compound indexes and TTL retention that cut query latency by 60% and slowed storage growth by 40%.
- Migrated all Pit Stop history (~15M records) into a multi-tenant schema using restartable Spring Batch jobs with checksum reconciliation, completing cutover with zero data loss and no downtime.
- Implemented global exception handling with @RestControllerAdvice, RFC 7807 ProblemDetail and custom Bean Validation, and structured JSON logging with MDC correlation IDs and PII masking, reducing incident triage from hours to under 15 minutes.
- Automated per-team Grafana 9 dashboards with Micrometer metrics for P90/P95/P99 latency, JVM health and VM cost, helping teams cut cloud spend by up to 20%, and built Slack alerting that reduced weekly L2 support tickets by 45%.
- Resolved recurring OutOfMemoryErrors through Eclipse MAT heap dump analysis and G1GC container-aware tuning, reducing peak heap usage by 40%, eliminating OOMKilled pod restarts and holding GC pauses under 200 ms.
- Orchestrated k6 runners as Kubernetes Jobs on a dedicated GKE node pool supporting 200+ concurrent tests and 50K+ virtual users, and deployed all services with multi-zone replicas and PodDisruptionBudgets for 99.9% availability.
- Built Jenkins pipelines with SonarQube and OWASP gates feeding Argo CD 2.6 GitOps and Helm 3 rolling deployments, moving releases from biweekly to 3 to 4 per week with zero downtime.
- Secured the platform with Spring Security 6 as an OAuth 2.0 resource server with OIDC SSO, JWT validation, four-tier RBAC, Workload Identity, Secret Manager and TLS, clearing security review with no high-severity findings.
- Maintained 85% backend coverage with Testcontainers, WireMock and Spring Cloud Contract, and built the React 18 frontend with React Query and lazy-loaded routes, tested with Jest, React Testing Library, Cypress and axe WCAG 2.1 AA checks, cutting initial page load by 40% and production defects by 35%.
- Owned on-call during peak retail events, authored RCAs, led code reviews and mentored two associate engineers in Agile/Scrum delivery.
- Built Pit Stop, a React 17 peak readiness portal on Lowe's Helix design system (100% design compliance) giving go/no-go deployment insights for 150+ APIs through 35+ Spring Boot REST APIs aggregating P90/P95/P99 metrics from InfluxDB, cutting deployment failures by 25%, speeding bottleneck resolution by 40% and supporting zero downtime through the holiday season.
- Designed pre-aggregated MongoDB schemas and time-windowed InfluxDB syncs with G1GC tuning for sub-second page loads and 50% lower sync heap usage, and implemented @ControllerAdvice exception handling, MDC request tracing and Spring Security SSO, cutting triage time by 40%.
- Set up Jenkins and Argo CD 2.2 pipelines with post-deployment smoke tests, reducing release time from ~2 hours to ~20 minutes, and maintained 75%+ coverage with JUnit 5, Mockito, Jest and React Testing Library, with every endpoint documented in OpenAPI 3.

TECHNICAL SKILLS

Programming Languages: Java (8, 11, 17, 21), Python, JavaScript, TypeScript, SQL, Go, Bash

Backend & Frameworks: Spring Boot, Spring WebFlux, Spring Batch, Spring Security, Spring Data JPA, Spring Data MongoDB, Hibernate, FastAPI, Microservices, Event-Driven Architecture, RESTful APIs, WebSockets

Frontend & Desktop: React, React Query, TypeScript, JavaFX, Responsive UI, WCAG 2.1 Accessibility

APIs & Integration: OpenAPI 3 (Swagger), API Versioning, Webhooks, SDK Development, FIX 4.4 (QuickFIX/J), HL7 FHIR R4, SMART on FHIR, OData, Idempotency, Rate Limiting (Bucket4j)

Messaging & Streaming: Apache Kafka, Amazon MSK, Event Streaming, Dead Letter Queues, Real-Time Market Data Feeds (TT, CQG, Rithmic, CTS)

Databases & Caching: PostgreSQL, SQL Server, MongoDB, InfluxDB, Redis, Caffeine, Flyway, Schema Design and Indexing

Cloud Platforms: AWS (EKS, EC2, S3, RDS, Lambda, Glue, API Gateway, MSK, KMS, Secrets Manager, Transfer Family, CloudWatch, CloudTrail, IAM), GCP (GKE, Cloud Storage, Cloud Logging, Secret Manager, Artifact Registry)

DevOps & Containerization: Docker, Kubernetes, Helm, KEDA, Rancher, Terraform, Jenkins, GitHub Actions, Argo CD (GitOps), Git, Blue-Green and Rolling Deployments, Infrastructure as Code

Observability & Performance: OpenTelemetry, Datadog, Grafana, Micrometer, SLF4J, Logback, JVM Tuning (G1GC, ZGC), Heap Dump Analysis (Eclipse MAT), Load Testing (k6, JMeter, Gatling, Locust)

Testing & Quality: JUnit 5, Mockito, Testcontainers, WireMock, Spring Cloud Contract, Pact, Pytest, Jest, React Testing Library, Cypress, Playwright, SonarQube

Security & Compliance: OAuth 2.0, OIDC, JWT, SSO, RBAC, Microsoft Entra ID, OWASP, Snyk, HIPAA, SOC 2, SOX/ITGC, CFTC and NFA Recordkeeping, KYC/AML, UK GDPR

Data Migration & ETL: ETL Pipeline Design, Spring Batch, Pandas, Data Reconciliation, Legacy Modernization (VAX to Microsoft Dynamics 365 F&O), D365 Data Management Framework

Domains: Capital Markets and FCM Clearing (SPAN Margin, Books and Records), Healthcare (EHR, Clinical Decision Support), Retail E-Commerce (Peak Readiness, Performance Engineering)

Leadership & Practices: Technical Leadership, System Design, Agile/Scrum, Jira, Confluence, Code Reviews, Mentoring, On-Call and RCA

CERTIFICATIONS

- AWS Certified Solutions Architect – Professional
- Certified Kubernetes Administrator (CKA)
- Project Management Professional (PMP)

EDUCATION

University of Southern California ***Los Angeles, CA, USA***

Master of Science (MS) in AI and Engineering Management *GPA: 4.0/4.0*